

# Farhan Kabir

## Senior Full-Stack Engineer | AI Agents, LLM Workflows & TypeScript Specialist

Rajshahi, Bangladesh | ✉ [farhankabir133@gmail.com](mailto:farhankabir133@gmail.com) | 📞 +8801783165726

🌐 [linkedin.com/in/farhankabir133](https://www.linkedin.com/in/farhankabir133) | 🐙 [github.com/farhankabir133](https://github.com/farhankabir133) | 🌐 [farhankabir.tech](https://farhankabir.tech)

### SUMMARY

Full-Stack Engineer with 4+ years of experience architecting production-grade web platforms, autonomous AI agents, and high-concurrency API microservices. Highly skilled in TypeScript, Node.js, Python (FastAPI, Django), React 19, Next.js, and PostgreSQL. Specialized in Model Context Protocol (MCP) implementations, LLM function orchestration, real-time WebSocket streaming, and automated background queues. Proven technical lead skilled in code reviews, system design, and AI-assisted workflows (Cursor, Copilot), driving 40% reductions in API latency and taking complex software products from idea to production with 99.9% uptime.

### EXPERIENCE

#### Lead Full Stack Engineer

Auto Spark 🌐

April 2024 – Present

Rajshahi, Bangladesh

- Direct technical architecture, lead code reviews, and establish engineering best practices for full-stack systems, serving as the primary architectural authority across client projects.
- Engineered high-concurrency Node.js and Python (FastAPI) API layers designed for 1,000+ concurrent WebSocket/REST connections, integrating LLM tool-calling and automated workflows.
- Designed relational PostgreSQL database schemas using Row Level Security (RLS) to process 500+ daily CRM leads across 10,000+ vehicle records with zero unauthorized data leaks.
- Cut average API response times by 40% (from 350ms to 210ms) via SQL query optimization, Redis caching, Nginx reverse proxying, and PM2/UptimeRobot health monitoring on Linux VPS.
- Integrated Sentry error-tracking and Dockerized GitHub Actions CI/CD pipelines, accelerating release cycles and eliminating production deployment regressions.

#### Full Stack Developer (Remote)

Hire My Tech 🌐

Aug 2022 – July 2023

Dhaka, Bangladesh

- Developed asynchronous backend microservices using Node.js, Express, and Python, integrating secure OAuth2/JWT token authentication and Redis rate-limiting.
- Architected resilient data-fetching pipelines with automated error boundary classification, retry logic, and centralized Sentry log streaming for distributed microservices.
- Collaborated directly with cross-functional product teams to translate complex business requirements into intuitive React/TypeScript user interfaces and RESTful endpoints.
- Containerized production environments with Docker and configured Jest/Pytest suites inside GitHub Actions CI/CD workflows, securing 99.9% system uptime.

### EDUCATION

#### B.Sc. in Computer Science and Engineering

Rajshahi University of Engineering and Technology (RUET)

2018 – 2022 | Score: 73.75%

### SKILLS

- **AI Agents & Workflows:** Model Context Protocol (MCP), LLM Agent Architectures, RAG Retrieval Systems, Tool Calling, Function Execution, OpenAI API, Gemini API, Claude 3.5.
- **Full-Stack & APIs:** TypeScript, JavaScript (ES6+), Node.js, Express, Python (FastAPI, Django, Flask), React 19, Next.js, RESTful APIs, WebSockets, GraphQL, CRM Integrations.
- **Databases & DevOps:** PostgreSQL, Supabase, Relational Schema Design, Row Level Security (RLS), MongoDB, Redis, Docker, Git, GitHub Actions, Linux (Ubuntu/Nginx), Vercel.
- **Testing & Observability:** Jest, Pytest, Postman, Sentry, PM2, UptimeRobot, System Architecture, Code Reviews, Customer Workflow Mapping, Agile Prototyping.

### KEY PROJECTS

#### 9Th-Grade AI – Exam Intelligence Platform | *Next.js 16, React 19, TypeScript, Tailwind CSS, Prisma, Groq API* [\[GitHub\]](#) [\[Live\]](#)

- Architected an open-source exam platform featuring BCS-accurate grading (+1/0.5), SM-2 spaced-repetition queues, a 60 FPS Canvas 2D simulation, and streaming bilingual AI tutors via Groq and Claude Sonnet 4.
- Engineered a Next.js 16 App Router system with PostgreSQL/Prisma ORM (20+ models), a 4-tier topic taxonomy (348 nodes), JWT HttpOnly auth, rate limiting, and 21 REST API routes backed by 61+ Vitest tests.

#### FarhanOS – Agentic Portfolio Workspace | *React 19, Vite 6, TypeScript, Three.js, Groq API, MCP*

[\[GitHub\]](#) [\[Live\]](#)

- Architected a full-stack windowed desktop-OS portfolio simulator implementing Model Context Protocol (MCP) primitives for dynamic tool discovery, context retrieval, and automated task execution.

- Built a floating AI assistant (/api/ask-twin, Llama 3.3 70B via Groq) supporting real-time WebSocket streaming, asynchronous state sync, and high-concurrency LLM inference orchestration.

**Vault – Secure Enterprise File Storage Platform** | Next.js 16, React 19, Express 4, TypeScript, PostgreSQL, AWS S3, Docker [\[GitHub\]](#) [\[Live\]](#)

- Architected a full-stack file storage platform featuring short-lived JWTs, rotating refresh tokens, double-submit CSRF defense, and non-blocking OTP email verification via SendGrid API.
- Engineered streaming multipart uploads with server-side magic-byte MIME validation, pluggable S3/local storage drivers, expiring public share links, and Dockerized CI/CD pipelines.

**FluentAI – AI Communication Coaching Platform** | Next.js 15, React 19, TypeScript, Tailwind CSS, Framer Motion [\[GitHub\]](#) [\[Live\]](#)

- Architected a high-performance, accessible landing platform featuring interactive AI coaching chat previews, multi-accent selection flows, and real-time analytics dashboards.
- Configured static site exports with Next.js 15 App Router, Framer Motion scroll reveals, custom Tailwind CSS v4 design tokens, and automated GitHub Pages deployment pipelines.

**The Ink Home – 3D Spatial Publishing Portal** | React 18, Vite, Three.js, Framer Motion, Node.js [\[GitHub\]](#) [\[Live\]](#)

- Architected a real-time server-side RSS ingestion pipeline mirroring a Medium publication into an immersive WebGL 3D reading experience.
- Designed a 4-tier cascading avatar-resolution system (direct fetch, rss2json, AllOrigins proxy, Unavatar) achieving reliable avatar delivery despite upstream CDN blocking.

**Auto Spark BD – Enterprise Dealership System** | React, TypeScript, PostgreSQL, Supabase, Python APIs [\[GitHub\]](#) [\[Live\]](#)

- Architected a production bilingual application managing 10,000+ inventory listings, featuring centralized PostgreSQL authorization and automated lead routing pipelines that cut manual processing time by 60%.
- Optimized asset loading, font subsetting, and Vite bundle splits, securing 90+ Lighthouse performance scores across all core production routes.

**BuildSafe Intelligence – Real-Time Decision Engine** | Next.js 15, Python APIs, TypeScript, Tailwind [\[GitHub\]](#) [\[Live\]](#)

- Developed a real-time decision engine that ingests atmospheric weather API streams and evaluates multi-variable operational risk matrices for automated alert distribution.
- Engineered a mobile-first frontend dashboard featuring interactive simulation controls, real-time data visualizations, and server-side evaluation engines.

## SOFTWARE SKILLS

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- **Development Tools & IDEs:** Cursor, Antigravity IDE, Visual Studio Code, Git, GitHub (Pull Shark Contributor), Docker, Postman, Insomnia.
- **Platforms & Hosting:** Vercel, DigitalOcean Droplets, Linux (Ubuntu/Debian Administration), macOS, Windows.
- **Design & Documentation:** Figma, UI/UX Wireframing, Responsive Design Systems, Technical Writing, System Architecture Diagrams, LaTeX.

## LEADERSHIP

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- **Technical Writer & Editor, Medium / The Ink Home:** Publish technical articles covering AI agent architectures, Model Context Protocol (MCP), full-stack performance tuning, and LLM tooling.
- **Organizer, Hult Prize RUET:** Coordinated cross-functional team workflows and sprint timelines during high-pressure innovation events.

## PUBLICATIONS

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- **Primary Author:** "AI-Driven Live Interview System for Real-Time Candidate Evaluation Using NLP and Computer Vision" – Int. Conf. on Computer and Information Technology (ICCIT), IEEE Proceedings. [\[Paper Link\]](#)
- **Primary Author:** "Emotion Detection From Textual Data Using Natural Language Processing and Machine Learning Techniques" – 4th Int. Conf. on Electrical, Computer and Communication Engineering (ECCE), 2025. [\[Paper Link\]](#)
- **Primary Author:** "Depression Detection From Social Media Textual Data Using Natural Language Processing and Machine Learning Techniques" – 26th Int. Conf. on Computer and Information Technology (ICCIT), 2023. [\[Paper Link\]](#)

## CERTIFICATIONS

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**Meta:** Full Stack Web Development | **Google:** Technical Writing | **IBM:** Cloud Computing

## AWARDS

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**Featured Author** – Medium (2025) | **Top 10 Finalist** – Hult Prize RUET (2023)

## LANGUAGES

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Bengali (Native) | English (Fluent) | Hindi (Conversational)